BASES DE DATOS AVANZADAS

Classroom: Bases de Datos Avanzadas

12 DE MAYO DE 2026

carlos.herrera@upa.edu.mx

https://github.com/ISC-UPA/2026-2-TIID5-DB2

Contenido

1. S01 2
    1.1. Referencias 2
    1.2. Technological Requirements 2
    1.1. Relaciones 3
        1NF 3
2. S02 6
3. S03 7

1. S01

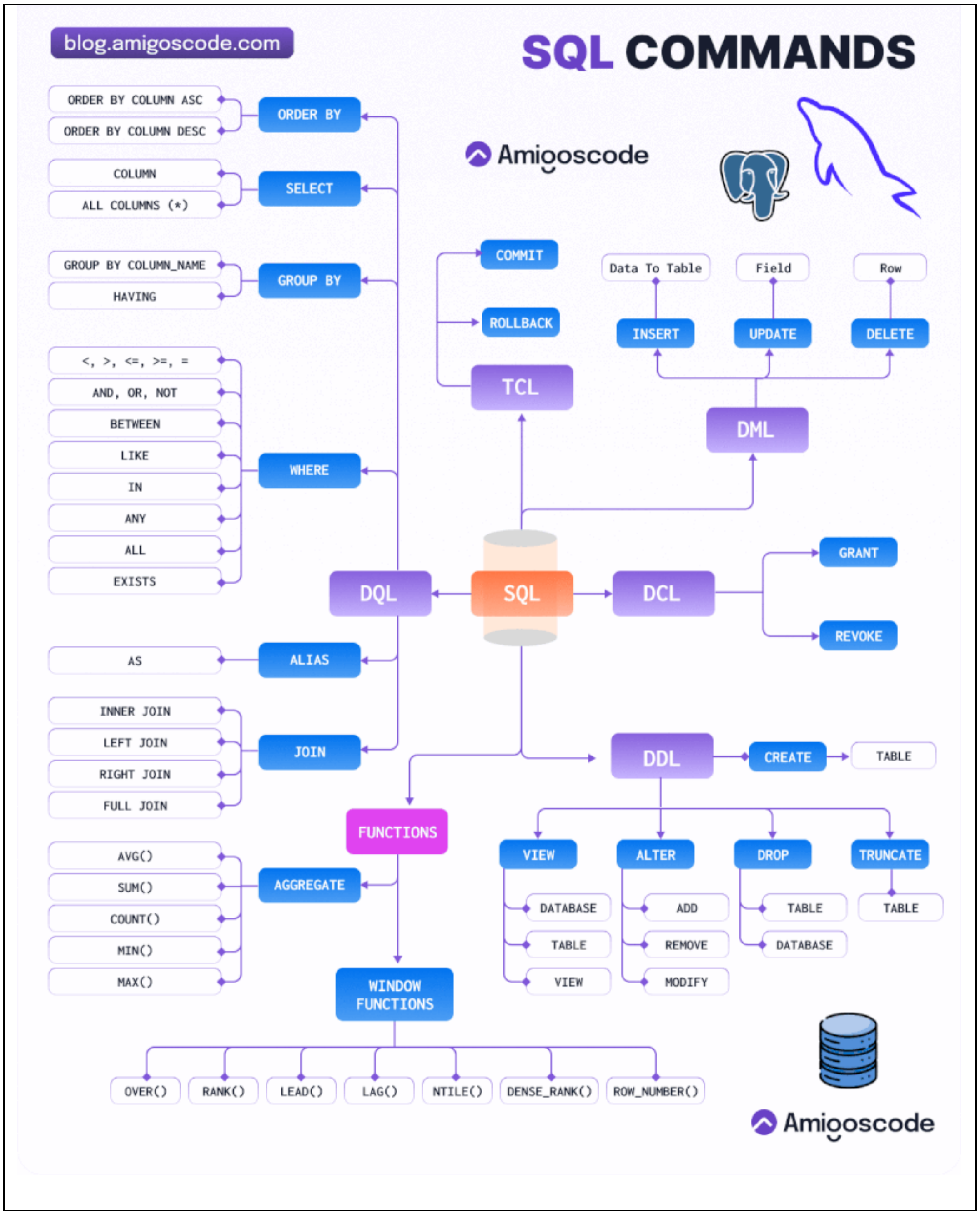
1.1. Referencias

https://www.w3schools.com/mysql
https://conclase.net/mysql

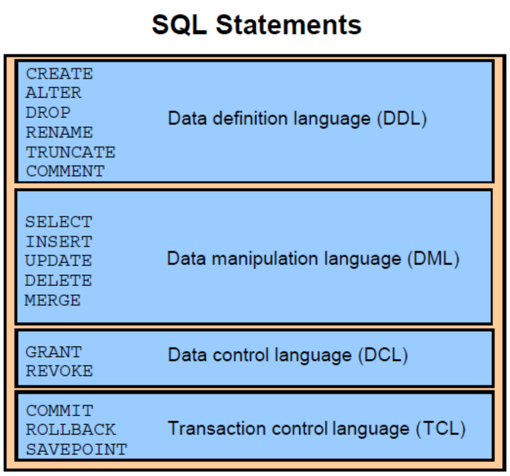
1.2. Technological Requirements

VS Code https://code.visualstudio.com/Download
Extensions: **Database Client JDBC**
Extensions: MongoDB for VS Code

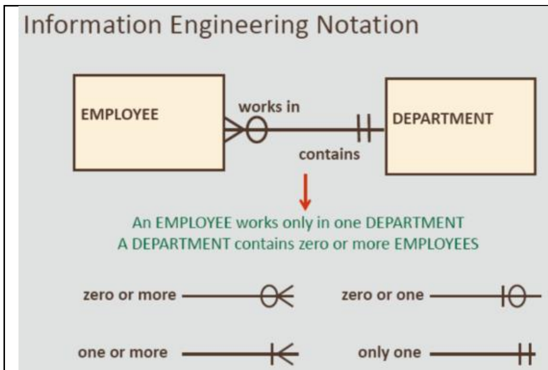
XAMPP **MongoDB** **PowerBI Desktop**

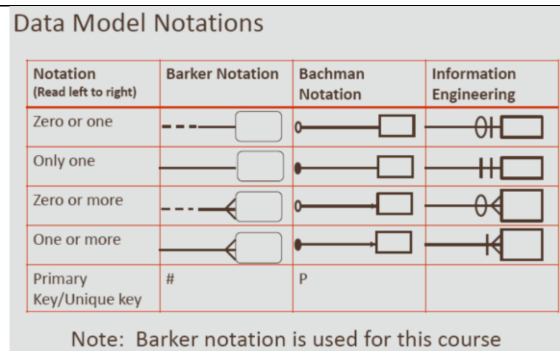


→



1.1. Relaciones



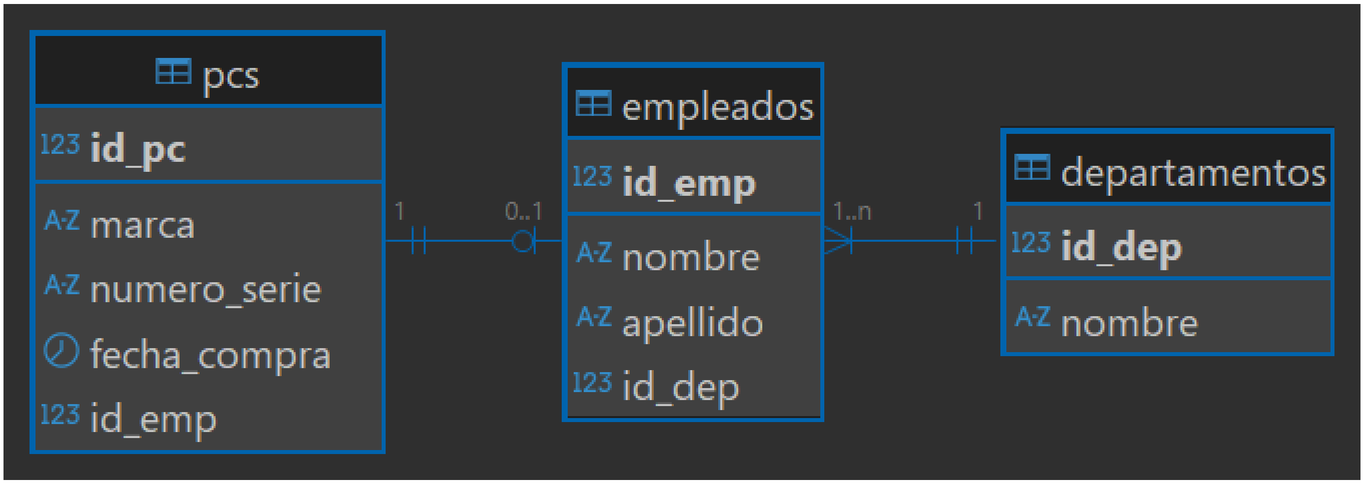


Normalization: Is the process of organizing the attributes and tables of a relational database to minimize redundancy. No duplicate content, Increase the integrity of data.

Rule	**Description**
First Normal Form (1NF)	All attributes must be single-valued. (no multi-valued = atomic)
Second Normal Form (2NF)	An attribute must be dependent on its entity's entire UID. No existen dependencias parciales
Third Normal Form (3NF)	No non-UID attributes can be dependent on another non-UID attribute. Ningún atributo no UID puede depender de otro atributo no UID. No existen dependencias transitivas.

1NF : If an attribute is multi-valued, create an additional entity and relate it to the original entity with a 1:M

→


pcs
id_pc
marca
numero_serie
fecha_compra
id_emp
1
0..1
empleados
id_emp
nombre
apellido
id_dep
1..n
1
departamentos
id_dep
nombre

→

```
-- Crear tabla de departmentos
CREATE TABLE departamentos (
    id_dep INT PRIMARY KEY AUTO_INCREMENT,
    nombre VARCHAR(50) UNIQUE NOT NULL
);

-- Crear tabla de empleados, debe pertenecer a un departamento
CREATE TABLE empleados (
    id_emp INT  AUTO_INCREMENT,
    nombre VARCHAR(50),
    apellido VARCHAR(50) NOT NULL,
    id_dep INT not null,
    CONSTRAINT pk_empleados PRIMARY KEY (id_emp),
    FOREIGN KEY (id_dep) REFERENCES departamentos(id_dep)
);

-- Crear una tabla de pcs,
-- cada pc puede estar asignada a un empleado
-- cada empleado debe tener una pc asignada.
CREATE TABLE pcs (
    id_pc INT PRIMARY KEY AUTO_INCREMENT,
    marca VARCHAR(50),
    numero_serie VARCHAR(20) NOT NULL,
    fecha_compra DATE,
    id_emp INT,
    UNIQUE INDEX unique_serie (numero_serie ASC) VISIBLE
);

create unique index unique_emp on pcs(id_emp);

alter table pcs
add constraint fk_pcs_emp foreign key (id_emp) references empleados(id_emp);

-- Insertar datos en la tabla de departamentos
INSERT INTO departamentos (nombre) VALUES ('Ventas');
INSERT INTO departamentos  VALUES (null, 'Informatica');
INSERT INTO departamentos  VALUES (null, 'Recursos Humanos');

-- Insertar datos en la tabla de empleados
INSERT INTO empleados (nombre, apellido, id_dep) VALUES ('Jesus', 'Pérez', 1);
INSERT INTO empleados  VALUES (null, 'María', 'Gómez', 2);
INSERT INTO empleados  VALUES (null, 'Jose', 'Franco', 2);

-- Insertar datos en la tabla de pcs
INSERT INTO pcs (marca, numero_serie, fecha_compra, id_emp) VALUES ('Dell', 'SN12345',
'2022-01-15', 1);
INSERT INTO pcs  VALUES (null, 'HP', 'SN54321', '2023-03-10', 2);
INSERT INTO pcs  VALUES (null, 'ASUS', 'SN67890', '2024-10-15', null); -- PC sin asignar

UPDATE pcs SET id_emp = 2 WHERE id_pc = 1;
-- Un Empleado una PC
```

→

Funciones: **select salary, codigo(salary) from employees;**
Procedimientos: **exec sp_calcularImpuesto @deptoid = 101;**

2. S02

Select	fields	
from	tables	
where	filters	logical operators: not and or
		relational operators: >,<,=,>=, =, <>, !=
		some words: like, between, in, is null, is not null
group by	functions: min, max, count, sum, avg, std	
having	filters with functions	
order by	fields [desc]	

Pattern Matching: LIKE Operator

- % denotes zero or more characters
- _ denotes one character

like 'A%' '_o%'

```
-- JOIN
SELECT table1.column, table2.column
FROM table1
        [NATURAL JOIN table2] |
        [JOIN table2 USING (column_name)] |
        [inner JOIN table2
                ON (table1.column_name = table2.column_name)] |
        [LEFT|RIGHT|FULL  OUTER JOIN table2
                ON (table1.column_name = table2.column_name)] |
        [CROSS JOIN table2];
```

→

3. S03

Funciones Escalares: String, Numeric, Date and Advanced (Flow Control)

https://www.w3schools.com/mysql/mysql_ref_functions.asp
https://conclase.net/mysql/curso/cap11
MySQL :: MySQL 8.0 Reference Manual :: 14 Functions and Operators
mariadb.com/docs/

Funciones de Cadena	Resultado	MySQL/Oracle
select ascii("A") Numero, char(65) as Caracter;	65 A	
FIELD("q", "s", "q", "l")	2	
FIND_IN_SET("q","s,q,l")	2	
FORMAT(250500.5634, 2)		
CONCAT('Hello', 'World')	HelloWorld	both
TRIM(' HOLA MUNDO ');	HOLA MUNDO	both
LENGTH('HelloWorld')	10	both
INSTR('HelloWorld', 'Wo')	6	both
INSTR('HelloWorld', 'o',1,2)	7	no
INSTR('HelloWorld', 'o',-1)	7	no
INSTR('HelloWorld', 'o',-1,2)	5	no
SUBSTR('HelloWorld',1,5)	Hello	both
SUBSTR('HelloWorld',6)	World	both
SUBSTR('HelloWorld', -3, 1)	r	both
where SUBSTR('Abel', -2, 1)='a'		
SUBSTR(SYSDATE, 8)	'YY' or 'YYYY'	both
LPAD(salary,10,'*')	*****24000	both
RPAD(salary, 10, '*')	24000*****	both
REPLACE('JACK and JUE','J','BL')	BLACK and BLUE	both
INITCAP('UPA')	Upa	No existe
lower('Upa')	upa	both
upper('Upa')	UPA	both
reverse('UPA')	APU	both
strcmp("carlo", "carla")	0 1 -1	
SUBSTRING_INDEX("upa.edu.mx", ".", 2)	upa.edu	

→

Funciones Numericas	Resultado	MySQL/Oracle
ROUND(45.926, 2)	45.93	both
ROUND(45.923,-1)	50	
TRUNC(45.926, 2)	45.92	
TRUNC(45.923,-1)	40	truncate
MOD(5, 3)	2	both
least(5,3,5,2,8)	2	both
greatest(5,3,5,2,8)	8	both
power(2,3)	8	both
exp(1)	2.7172	both
abs(-5)	5	both
ln(10)	2.3025	both
log(10,100)	2	both
sign(12)	1	both
sign(0)	0	
sign(-12)	-1	

```
Funciones de Fechas                          Resultado
current_date(), sysdate()
user(), current_user(), session_user()
LAST_DAY("2017-06-15");                                  30/06/2017
ADDDATE("2026-02-25", INTERVAL 10 DAY)                   07/03/2026
SUBDATE("2017-06-15", INTERVAL 10 DAY)                   05/06/2017
DATE("2017-06-15")                           Convierte a fecha
DATEDIFF("2017-06-25", "2017-06-15")         Regresa Numero de dias
DATE_FORMAT("2017-06-15", "%Y")                                2017
year("2017-06-15")  , month(), day()                           2017
DAYNAME("2017-06-15")                        Thursday
DAYOFWEEK(CURDATE())                         1=Sunday. .7=Saturday
WEEKDAY(sysdate())                           0=Monday. .6=Sunday
EXTRACT(MONTH FROM "2017-06-15")                  both   1..12
MONTHNAME("2017-06-15")                      June
STR_TO_DATE("August 10 2017", "%M %d %Y");
```

→

Format	Description
%a	Abbreviated weekday name (Sun to Sat)
%b	Abbreviated month name (Jan to Dec)
%c	Numeric month name (0 to 12)
%D	Day of the month as a numeric value, followed by suffix (1st, 2nd, 3rd, ...)
%d	Day of the month as a numeric value (01 to 31)
%e	Day of the month as a numeric value (0 to 31)
%H	Hour (00 to 23)
%h	Hour (00 to 12)
%i	Minutes (00 to 59)
%j	Day of the year (001 to 366)
%M	Month name in full (January to December)
%m	Month name as a numeric value (00 to 12)
%p	AM or PM
%r	Time in 12 hour AM or PM format (hh:mm:ss AM/PM)
%s	Seconds (00 to 59)
%T	Time in 24 hour format (hh:mm:ss)
%u	Week where Monday is the first day of the week (00 to 53)
%V	Week where Sunday is the first day of the week (01 to 53). Used with %X
%v	Week where Monday is the first day of the week (01 to 53). Used with %x
%W	Weekday name in full (Sunday to Saturday)
%w	Day of the week where Sunday=0 and Saturday=6
%X	Year for the week where Sunday is the first day of the week. Used with %V
%x	Year for the week where Monday is the first day of the week. Used with %v
%Y	Year as a numeric, 4-digit value
%y	Year as a numeric, 2-digit value

→